



Trusting AI to detect AI? A systematic evaluation of the reliability and robustness of current AIGC detection tools for student academic work

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ARTICLE INFO

Keywords:

Academic integrity
AIGC detection
Large language models
Engineering education
Assessment reform
Higher education

ABSTRACT

The explosive proliferation of Generative Artificial Intelligence (GenAI) has posed an unprecedented existential challenge to traditional evaluation systems in higher education. In response to this wave, the higher education sector is undergoing a profound transformation, shifting from a focus on technological containment toward an assessment governance paradigm. Understanding the characteristics of detectors is the foundational prerequisite for utilizing such tools to optimize educational assessment; however, their effectiveness within authentic educational contexts remains insufficiently explored. To address this research gap, the study constructs three large-scale ecological datasets—StuTask, StuThesis, and DataCode—comprising over 280,000 authentic samples of student coursework, academic theses, and engineering code. A systematic evaluation was conducted on 13 mainstream detectors, covering both commercial and open-source models, across multiple dimensions including overall performance, task complexity, disciplinary variations, and adversarial robustness. The results indicate that while detectors achieve acceptable performance on long-form theses, they exhibit systematic failures in engineering code and short-form coursework tasks. Due to the formulaic nature of technical writing, STEM disciplines are subject to significant algorithmic bias. Furthermore, robustness tests reveal the extreme vulnerability of current detection tools, as a hybrid editing strategy can enable up to 88% of AI-generated content to evade detection successfully. These findings suggest that existing detection technologies are inadequate to support high-stakes educational assessments. In the future educational trajectory of ‘embracing AI,’ AIGC detectors should function as reference metrics within the assessment system, serving to quantify the depth of human–AI collaboration across distinct disciplinary logical frameworks. Detection technology must evolve toward the optimization of ‘logical innovation recognition,’ thereby establishing a robust academic integrity defense line that is truly resilient within the future ecosystem of human–AI symbiosis.

1. Introduction

The rapid advancement of Large Language Models (LLMs) represented by ChatGPT and Gemini has profoundly reshaped the ways in which students engage in academic writing and polishing, fundamentally transforming the teaching and assessment paradigms of higher education (Kasneci et al., 2023; Perkins, Roe, Vu et al., 2024). Leveraging its robust natural language processing (NLP) and

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<https://doi.org/10.1016/j.compedu.2026.105616>

Received 1 December 2025; Received in revised form 9 February 2026; Accepted 14 March 2026

Available online 19 March 2026

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generation capabilities, Artificial Intelligence Generated Content (AIGC) technology can not only assist students in brainstorming and language polishing but also generate academically sound texts—characterized by grammatical accuracy, logical coherence, and semantic richness—in a matter of seconds (Alfredo et al., 2024; Baidoo-Anu & Ansah, 2023). For educators and students in higher education, this constitutes not only an unprecedented opportunity to enhance learning efficiency but also a profound challenge to the traditional academic assessment system.

However, this technological convenience has also aroused profound concerns about an “academic integrity crisis” within the global education community (Fleckenstein et al., 2024; Yan et al., 2024). As students increasingly rely on AI-generated content for coursework, theses, and even programming assignments, how to define authorship and uphold the fairness of assessments has become a pressing issue that demands immediate attention (Cotton et al., 2024; Sun et al., 2026). If it remains impossible to effectively distinguish between work independently completed by students and AI-generated content, this will not only devalue academic degrees but also potentially lead to the systematic failure of educational institutions in grading and feedback provision (White, 2023).

Initially, this technological convenience has also aroused profound concerns about an “academic integrity crisis” within the global education community (Fleckenstein et al., 2024; Yan et al., 2024). In response, the academic and industrial sectors have rapidly rolled out a suite of AIGC detectors (Cui & Zhang, 2025; Pan et al., 2024), seeking to regain the initiative in this “cat-and-mouse game”. These tools—including commercial solutions like GPTZero and Originality.ai, as well as open-source models developed by the academic community, such as Fast-Detect GPT—claim to accurately identify traces of AI generation by analyzing statistical features of texts, such as perplexity and burstiness (Tian et al., 2023). Many higher education institutions have also begun integrating these tools into their daily assignment grading workflows, pinning their hopes on technological means to curb academic misconduct.

However, as higher education shifts from “vigilance” toward “collaboration and embrace” regarding AI tools, the traditional practice of judging and utilizing detectors solely based on effectiveness has gradually failed to adapt to the developmental requirements of the era of human–AI symbiosis. Under the pedagogical vision of “embracing AI”, detectors should not merely serve as “police” maintaining integrity. Still, they should instead become a kind of diagnostic indicator for evaluating the degree of AI penetration, understanding technological boundaries, and ensuring assessment fairness (Gulumbe et al., 2025). This shift necessitates a transition from traditional document-level binary classification toward more granular identification tasks, such as automatic boundary detection, which aims to pinpoint the precise transition points between human and AI contributions in student writing (Zeng, Sha et al., 2024). Under such mandates, is our current conceptualization of detectors robust enough to justify their deployment as reference indicators for higher education assessment?

Firstly, the inauthenticity of evaluation data: many existing assessment studies rely on Wikipedia entries, news articles, or Reddit forum posts as “human-written” samples (Sadasivan et al., 2023). However, student academic writing possesses distinct stylistic features, reasoning logics, and disciplinary norms, and texts from general domains cannot accurately reflect students’ authentic linguistic habits in coursework and theses. Such data bias may lead to a significant performance degradation of detectors when confronted with real student assignments, rendering their utility as reference indicators for assessment highly questionable.

Secondly, current evaluations primarily focus on pure text writing in the humanities and social sciences (HSS), with few studies examining code generation and technical problem-solving in science, technology, engineering, and mathematics (STEM) fields (Liu et al., 2023). With the widespread adoption of AI-assisted programming in computer science and engineering disciplines, evaluating the performance of detectors in code-related scenarios has become an urgent priority (Chang et al., 2024). Since linguistic styles and structural patterns vary drastically across different disciplines (e.g., literature and engineering), a nuanced understanding of detector performance on highly structured texts constitutes a fundamental prerequisite for constructing a fair and interdisciplinary AI evaluation framework.

Finally, in the current learning and working environment that advocates for ‘human–AI collaboration’, students’ creative paradigms increasingly involve manual intervention at both ‘ends’ of the creative process. Specifically, based on self-conceived frameworks, students engage in the active editing of raw LLM outputs—performing human–AI collaborative polishing through synonym replacement and sentence restructuring—to more effectively articulate their personal viewpoints. Such collaborative paradigms create complex “hybrid texts” where human intervention often obscures machine fingerprints, making the identification of individual AI-generated sentences significantly more challenging than detecting monolithic, unedited AI outputs (Zeng, Liu et al., 2024). Recent studies have shown that current detectors may exhibit biases against non-native English speakers and are highly vulnerable to being deceived by simple paraphrasing techniques (Liang et al., 2023; Weber-Wulff et al., 2023). Meanwhile, detection tools often misclassify complex human-written content as AI-generated, and such false positives may lead to serious ethical and disciplinary issues, undermining students’ trust in assessments and compromising fairness (Chen et al., 2025; Cui & Zhang, 2025). To date, no studies have systematically compared the performance of various detectors or established standardized evaluation criteria, leading to polarized extremes in detector-assisted educational assessment.

To address the aforementioned research gaps, this study constructs three representative datasets—*StuTask*, *StuThesis*, and *DataCode*—derived from authentic historical student assignment archives collected at a comprehensive university between 2016 and 2021, a period preceding the widespread adoption of generative AI tools. Using a controlled parallel corpus construction strategy, each dataset consists of paired human-written assignments and their corresponding AI-generated counterparts, resulting in a total of over 280,000 data instances. Based on these datasets, we evaluate 13 representative AIGC detectors, including 7 open-source models and 6 widely deployed commercial tools, to systematically examine detection performance across task types and disciplines. Through this design, the study aims to provide a comprehensive assessment of the effectiveness, limitations, and robustness of current AIGC detectors in authentic academic settings. In summary, this study addresses the following four research questions through large-scale empirical analysis:

- **RQ1:** How effective are existing detectors in detecting LLM-generated content in student academic writing?
- **RQ2:** What factors of the generated content affect the detection performance?
- **RQ3:** How do different disciplines influence the detectability of generated content?
- **RQ4:** How robust are these detectors when the LLM-generated data is slightly modified?

Although this study evaluates the performance of multiple AIGC detection tools across a wide range of tasks, disciplines, and realistic usage scenarios, its primary purpose is not to benchmark detectors in a purely technical sense. Rather, against the backdrop of the necessity to cultivate new-era talents capable of human–AI collaboration, the reform of higher education assessment inevitably necessitates a shift toward the critical use of detector-assisted evaluation. This work aims to critically examine whether current detection tools are sufficiently reliable, robust, and fair to justify their deployment in high-stakes educational assessment and institutional decision-making. By situating detector performance within authentic student activities—including coursework, theses, and engineering code—and by explicitly modeling disciplinary variation and realistic adversarial behaviors, our evaluation moves beyond accuracy comparisons to deeply interrogate the practical trustworthiness of detection-based paradigms. Consequently, this study reframes AIGC detection from a problem of technical feasibility to one of educational appropriateness, offering empirical evidence to reveal the adaptability of AI detection across diverse disciplines and tasks and the extent to which it poses potential threats to assessment fairness, thereby providing a reference for the application of AI detection in educational evaluation.

2. Literature review

2.1. The impact of artificial intelligence on higher education assessment

Against the backdrop of the deep integration of Generative Artificial Intelligence (GenAI) into higher education, assessment paradigms are undergoing a transformation in tandem with the realization that ‘society requires talents capable of leveraging AI to enhance their work’. Higher education assessment is shifting from the ‘compliance-driven prevention’ characteristic of the period of high resistance to AI toward a reflection on assessment priorities and methodologies. This transformation originates from the evolving conceptualization of AI-assisted assignment completion.

Academic integrity has long been regarded as the core cornerstone of higher education. The International Centre for Academic Integrity (ICAI) defines it as a commitment to six fundamental values: honesty, trust, fairness, respect, responsibility, and courage (ICAI, 2021). The imperative for ‘integrity’ has not been marginalized by the emergence of Large Language Models (LLMs). In recent years, a growing body of scholars has explored the impact of Generative Artificial Intelligence (GenAI) on the concept of academic integrity and its subsequent reconstruction. Cotton et al. (2024) argue that the emergence of tools like ChatGPT has blurred the line between “assistive tools” and “cheating”. By instantly generating high-quality, plagiarism-free text, GenAI has transformed traditional academic misconduct into a more undetectable form of “AI-assisted cheating”. This challenge lies not only in the increased difficulty of detection but also in the shifting nature of the assessment object. White (2023) argues that this technological transformation is compelling institutions to re-examine the definition of ‘authorship’. When finally submitted, artifacts are no longer reliable proxies for students’ authentic abilities; assessment paradigms must undergo a fundamental shift. Whereas students previously incurred high costs to outsource assignments to third-party services, free or low-cost generative AI services are now readily accessible. Consequently, students may easily gravitate toward a consequentialist approach to coursework (Zalta et al., 2003), prioritizing task completion over institutional academic norms, which poses a significant threat to universities and their evaluative processes. The concept of ‘Cognitive Offloading’ serves as a reminder that traditional assessments often focus exclusively on ‘offline capabilities’, thereby neglecting actual efficacy under human–AI collaboration (Dawson, 2020). Accordingly, researchers are increasingly advocating for a shift in the evaluative focus from static manuscript outputs to the dynamic tracking of thinking trajectories. The once purely negative stance toward AI’s role in generating student work is thus evolving into a more nuanced and diverse perspective through critical reflection.

The responses of higher education institutions to AIGC can be delineated into three paradigms: ‘AI-restrictive’, ‘AI-supportive’, and ‘AI-embracing’. Novian et al. (2024) advocate for the philosophy of ‘embracing rather than avoiding’, contending that GenAI should be conceptualized as a catalyst for reconstructing personalized learning and research innovation, rather than a mere threat. Concurrently, Gulumbe et al. (2025) underscores the necessity for academic publishers and universities to seek a dynamic balance between innovation and integrity. Prior research has frequently leaned toward the ‘restrictive’ paradigm, attempting to re-establish technical barriers through AIGC detectors. However, Kofinas et al. (2025) demonstrates that even within the framework of ‘authentic assessment’—a method once heavily relied upon—experts fail to reliably differentiate between AI-generated and human-written artifacts; thus, the higher education sector cannot depend exclusively on authentic assessment to safeguard academic integrity. The erosion of these technical defenses is necessitating a paradigm shift toward ‘AI-embracing’, where AI literacy is integrated into evaluative metrics. Furthermore, amidst the AI surge, the educational community has increasingly recognized that static product-based evaluations are inadequate for reflecting student learning outcomes. Within the learning sciences, learning is defined by change—including transformations in psychological representations (Perret-Clermont, 1980), the shift from scaffolded performance to independent mastery (Vygotsky, 1978), and the process of cultural adaptation (Lave & Wenger, 1991). While AI excels at simulating outcomes, it cannot replicate the cognitive journey of challenges and insights inherent in the learning process; consequently, assessment must aim to make this process ‘visible’. This transition is manifested in the rising prominence of ‘process-oriented assessment’ (Swiecki et al., 2022). Moreover, Bearman et al. (2023) posits that the most essential skill in the AI age is ‘evaluative judgment’—the capacity to discern and calibrate the quality of work. This competency is particularly crucial in human–AI collaboration, where assessment evolves from a unidirectional grading mechanism into a collaborative inquiry between educators

and students regarding the nature of high-quality work. The [TEQSA \(2023\)](#) policy framework further specifies that assessment tasks should be engineered to prompt students to critically analyze the role of AI within disciplinary contexts, rather than merely banning its application.

With the transformation of assessment paradigms, how to leverage existing tools to support the holistic design of higher education assessment has emerged as a critical research question. Most existing assessment studies rely heavily on general texts such as news articles or encyclopedias ([Sadasiyan et al., 2023](#)), lacking large-scale empirical data based on authentic student assignments. Furthermore, [Liang et al. \(2023\)](#) pointed out that existing studies have barely focused on the potential biases of detection algorithms against non-native speakers, and have paid insufficient attention to the increasingly prevalent issue of code generation in engineering disciplines. These research gaps indicate an urgent need for a systematic evaluation that covers multiple disciplines and genres and is based on authentic educational scenarios, so as to clarify the actual effectiveness of detection tools in safeguarding academic integrity.

2.2. Existing detectors and their accuracy assessment

To address the proliferation of AI-generated content (AIGC) in educational settings, the academic and industrial sectors have developed a variety of detection tools. Based on differences in technical transparency and access methods, existing detectors are primarily categorized into two types: open-source detectors built on public algorithms and commercial detectors based on proprietary services.

Open-source detectors typically emerge as products of academic research, with their core advantage lying in algorithmic transparency, which allows researchers to conduct in-depth analysis of their decision-making logic. Early detection tools were mainly based on supervised learning. For instance, the GPT-2 Output Detector released by OpenAI is fine-tuned on the RoBERTa model, distinguishing between human-written and AI-generated text by learning from a large number of text pairs ([Solaiman et al., 2019](#)). Similarly, models such as RoBERTa-QA and ArguGPT focus on specific text genres. Among them, ArguGPT, developed by [Liu et al. \(2023\)](#), has been specifically optimized for argumentative essays, aiming to improve detection rates by capturing the rigid patterns of AI in argumentative logic.

However, as the iteration speed of Large Language Models (LLMs) outpaces the update rate of training data, supervised learning-based methods face a severe bottleneck in generalization ability. To address this issue, research has shifted toward zero-shot detection techniques, leveraging the intrinsic probabilistic and statistical characteristics of LLMs—such as “Perplexity” and “Burstiness”—for judgment without the need for specialized training data ([Mitchell et al., 2023](#)). DetectGPT pioneered the use of log probability curvature to identify AI-generated text, arguing that text produced by large language models (LLMs) typically lies in the negative region of the probability curvature ([Mitchell et al., 2023](#)). Building on this foundation, [Bao et al. \(2023\)](#) proposed Fast-DetectGPT, which significantly improved detection speed and accuracy by introducing conditional probability curvature, emerging as one of the state-of-the-art (SOTA) models in the open-source domain. Furthermore, models such as MAGE and RADAR have incorporated adversarial learning mechanisms, aiming to enhance model robustness through the interplay between generation and detection ([Hu et al., 2023](#)).

Unlike open-source models, commercial detectors such as GPTZero, Originality.ai, CopyLeaks, and Scribbr are typically provided as API or web-based services, which are widely integrated into Learning Management Systems (LMS) like Canvas and Blackboard. These tools are often regarded as “black boxes” because their specific training data and algorithmic weights are not publicly available.

GPTZero is one of the most widely used tools in the educational sector currently. It primarily relies on metrics such as “Perplexity” and “Burstiness” to measure text complexity, claiming to effectively distinguish between the randomness of human writing and the fluency of AI-generated text ([Pauillet et al., 2025](#)). Originality.ai, by contrast, positions itself as an all-in-one detector specifically designed for ChatGPT and GPT-4. In addition to identifying AI-generated content, it integrates traditional plagiarism detection functionality, making it widely adopted by numerous universities for the comprehensive review of term papers ([Elkhatat, 2023](#)). In addition, Writefull and Writer.com, originally developed as academic writing assistance tools, have now added AI detection features, aiming to assist students in self-checking or teachers in preliminary screening. Despite claiming extremely high accuracy rates (typically advertised as >95%), these commercial tools often leave educators unable to explain their decision-making basis due to the opacity of their algorithms, leading to frequent grading disputes ([Perkins, Roe, Postma et al., 2024](#)).

Despite the proliferation of AIGC detection tools claiming an accuracy rate of up to 99% on the market, independent academic studies have shown that their performance in real educational settings falls far short of expectations. Existing accuracy research primarily focuses on three aspects: benchmark testing, algorithmic bias, and adversarial robustness, while also exposing significant gaps in the current research landscape.

In controlled experimental environments, some detectors have demonstrated a certain degree of discriminative capability. For instance, [Liu et al. \(2023\)](#) found in their evaluation of ArguGPT that models optimized for specific genres (e.g., argumentative essays) can capture the logical stereotypes commonly present in AI-generated text. However, when these tools are confronted with the complex and variable student assignments in real-world scenarios, their performance often declines sharply. [Weber-Wulff et al. \(2023\)](#) conducted a landmark cross-sectional evaluation of 14 mainstream detectors, including GPTZero, Turnitin, and Scribbr. The results showed that none of the tools achieved a sufficiently high accuracy standard to support “academic misconduct penalties”. In particular, a study by [Elkhatat \(2023\)](#) points out that the accuracy of commercial detectors is highly unstable: they perform moderately well when detecting content generated by GPT-3.5, but their detection rates decrease significantly when confronted with text produced by more advanced models such as GPT-4. Furthermore, [Gotoman et al. \(2025\)](#) found that paid tools generally

perform slightly better than free counterparts, yet the issue of high false positive rates remains a pervasive concern that cannot be overlooked.

The most concerning flaw of detection tools lies in their high false positive rates, which have triggered a severe ethical crisis in higher education. Dalalah and Dalalah (2023) warn that many detectors operate based on statistical metrics such as “Perplexity” and “Burstiness”, leading to a high risk of misclassifying fully human-written texts—characterized by rigorous writing, adherence to academic templates, or high vocabulary repetition—as AI-generated.

What is more serious is that such misclassifications are not randomly distributed but exhibit systematic group bias. A study conducted by Liang et al. (2023) at Stanford University revealed a significant bias of detectors against non-native English speakers (NNES). The study found that due to non-native English-speaking students’ typically lower vocabulary richness and relatively simple sentence structures (i.e., low perplexity), their risk of being falsely flagged by tools like GPTZero is as high as 61%, whereas the false positive rate for native speakers under the same conditions is nearly zero. This indicates that existing detection algorithms may inadvertently penalize student groups with weaker language proficiency, which seriously violates the fairness principle of educational assessment (Walters, 2023).

Insufficient robustness constitutes another major weakness of existing detectors. Studies by Sadasivan et al. (2023) and Wolff (2020) have demonstrated that AIGC detectors are highly vulnerable to adversarial attacks. While attacks in the field of computer science typically involve complex algorithmic perturbations, students in educational settings mostly adopt simple “manual paraphrasing” strategies.

Empirical research by Krishna et al. (2023) has shown that merely using tools like Quillbot for paraphrasing (i.e., synonym replacement) can cause the detection accuracy of GPTZero and Originality.ai to plummet to the level of random guessing. Furthermore, Coz et al. (2023) found that by adding simple instructions such as “Please write like a human being” or “Use more colloquial expressions” to the prompt, the generated text can easily bypass most existing open-source detectors. This “easily breached” vulnerability exposes a fundamental flaw in current systems: they primarily rely on superficial statistical features rather than deep semantic understanding. This means students can evade detection through minimal-cost modifications.

Although it has become quite common for science and engineering students to use AI for code generation to complete their assignments, there has been scarce research on the effectiveness of detectors in code-related scenarios. Preliminary explorations by JianWang et al. (2024) have indicated that existing text detectors, such as Fast-DetectGPT, can hardly be directly applied to the code domain. This is because programming languages are highly structured and repetitive, which renders the “perplexity” metric ineffective. Currently, the evaluation of detectors against engineering-specific datasets such as Stack Overflow remains nearly non-existent. This leaves engineering educators without effective tools to address AI-assisted programming.

In summary, to accurately assess the practical effectiveness of AIGC detection tools in higher education, it is imperative to construct a large-scale benchmark that includes authentic student assignments, covers all disciplines (including code), and accounts for real-world modification behaviors. This constitutes the core motivation behind the construction of the StuTask, StuThesis, and DataCode datasets in this study, as well as the subsequent empirical analysis.

3. Methodology

To systematically evaluate the effectiveness of AIGC detection tools in real higher education scenarios, this study designed a multi-stage empirical evaluation framework (as shown in Fig. 1). This framework covers the entire process, ranging from the collection of authentic educational data, generation of parallel corpora, large-scale automated detection to multi-dimensional performance analysis.

At the outset, we constructed three datasets—StuTask, StuThesis, and DataCode—representing three core cognitive tasks: knowledge comprehension, academic synthesis, and technical application. Subsequently, 13 mainstream detection tools (including both open-source and commercial models) were used to conduct benchmark testing on the aforementioned datasets. Finally, we performed in-depth analysis of the detection results from four dimensions: overall effectiveness (RQ1), the impact of content features (RQ2), disciplinary differences (RQ3), and adversarial robustness (RQ4), aiming to provide a scientific basis for the formulation of academic integrity policies.

3.1. Dataset construction

Existing detector evaluations struggle to reflect the complexity and particularity of student writing in higher education. To address this gap, this study constructed a corpus representing three typical forms of student assignments in higher education—short course assignments, long academic theses, and engineering code—based on authentic archived student assignments from the pre-AI era and data from online programming communities. To eliminate the interference of topic bias on detection performance, all datasets adopted a parallel corpus generation strategy: for each rigorously screened human-written text, we used OpenAI’s GPT-3.5-turbo-0125 model to generate a corresponding AI version under standardized prompt conditions.

To systematically evaluate the reliability of AIGC detection tools across different academic backgrounds and writing genres, this study constructed three independent datasets—StuTask, StuThesis, and DataCode. These datasets represent three typical forms of student assignments in higher education, namely short course assignments and exams, long academic theses, and code-based engineering problem-solving.

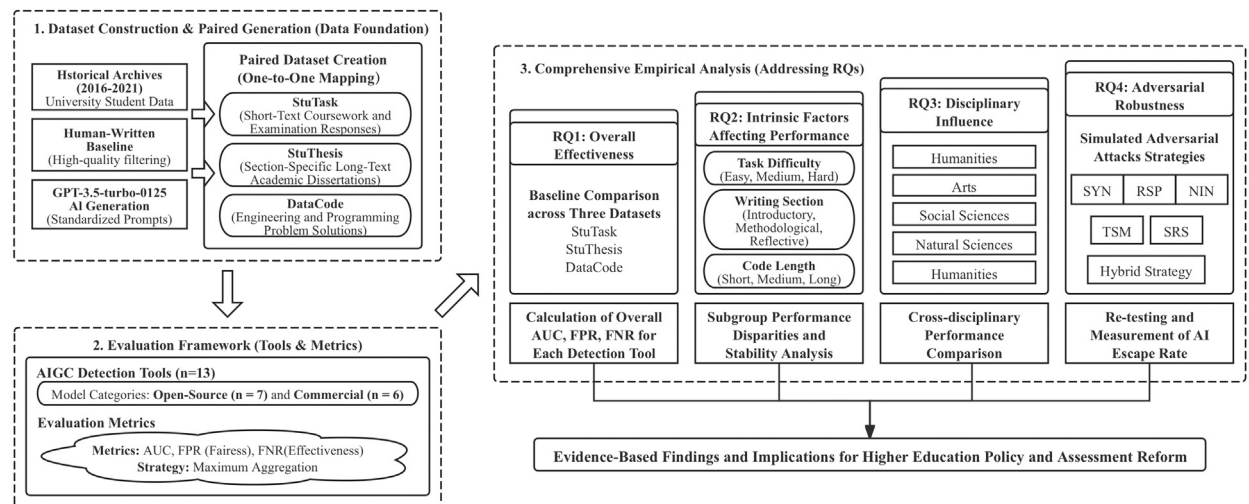


Fig. 1. The overall framework of the empirical evaluation workflow.

3.1.1. StuTask: Coursework and examination dataset

The StuTask dataset is designed to simulate basic academic tasks in daily teaching, focusing on assessing students' ability to explain concepts and conduct short-form reasoning. The data is sourced from archived regular course assignments and final exam scripts of students at our university between 2016 and 2021, covering all major disciplinary areas such as humanities, arts, social sciences, natural sciences, and engineering. To ensure that the human-written texts serving as the benchmark exhibit high academic standardization, we implemented a rigorous screening process, retaining only high-scoring samples with a score rate exceeding 85%. For each original question, we constructed the following prompt to generate AI responses:

"Please answer the following question as accurately and comprehensively as possible. Write your response in clear academic English, following the reasoning expected in a student assignment. Question: [Insert Question]."

Based on this, we generated multiple parallel human-AI pairs for each task. Ultimately, the StuTask dataset contains 26,037 pairs of samples (totaling 52,074 individual data entries). The average length of human-written answers is 269 words, while the average length of AI-generated answers is slightly higher, at 342 words.

3.1.2. StuThesis: Academic writing dataset

To address high-order academic writing scenarios, we constructed the StuThesis dataset, focusing on evaluating the performance of detection tools in long-form academic argumentation and synthetic writing. The data is derived from undergraduate and postgraduate theses collected between 2016 and 2021. Considering students' varying tendencies to use AI across different chapters, we adopted a stratified sampling strategy: approximately 80% of the samples were selected from the introduction, literature review, and discussion sections to cover high-frequency AI-assisted writing areas; the remaining 20% were sourced from the methodology and results sections to maintain diversity in the technical descriptive styles of the data. To generate contextually coherent AI texts, we input the thesis title and outline as background information into the model, using the following prompt:

"You are writing an academic paper. Based on the following topic and writing outline, generate a well-structured section that corresponds to the part indicated. Maintain a scholarly tone and coherence suitable for university-level research writing. Topic: [Insert Thesis Title]. Writing Plan: [Insert Thesis Outline]. Section Requested: [e.g., Literature Review]."

Ultimately, this dataset contains 74,581 pairs of samples, with the average lengths of human-written and AI-generated texts being 817 words and 984 words, respectively. It ranks among the largest student academic thesis datasets currently available.

3.1.3. DataCode: Engineering and programming dataset

For engineering disciplines, writing code to solve practical problems is a core component of higher education. The construction of the DataCode dataset is intended to address the lack of systematic evaluation resources for AIGC detection in programming assignments. We collected data from the Stack Overflow platform within the Stack Exchange network, covering the period from September 2021 to November 2022. This time window precedes the widespread adoption of large-scale generative code assistants in educational and public Q&A contexts, thereby reducing the likelihood of pervasive AI-generated content in the source data.

To ensure that the selected samples reasonably approximate human-authored code relevant to university-level instruction, we adopted a multi-stage validation process. First, we restricted candidate samples to answers that were explicitly accepted by the

Table 1

Overview of the 13 AIGC detectors evaluated in this study.

Category	Detector	Detectable models					Interfaces		Input length
		GPT-2	GPT-3	ChatGPT	Llama	Unknown	Website	API	
Open-source	GPT2-Detector (Veyseh et al., 2021)	✓						✓	<512 tokens
	RoBERTa-QA (Guo et al., 2023)			✓			✓	✓	<512 tokens
	ArguGPT (Liu et al., 2023)	✓	✓	✓			✓	✓	<512 tokens
	MAGE (Li et al., 2024)	✓	✓	✓			✓	✓	<2048 tokens
	RADAR (Hu et al., 2023)	✓	✓	✓			✓	✓	<512 tokens
	FastDetectGPT (Bao et al., 2023)	✓	✓	✓				✓	<512 tokens
	Multiscale (Tian et al., 2023)	✓	✓	✓			✓	✓	<512 tokens
Commercial	Crossplag.com (Crossplag, 2023)			✓			✓	✓	<3k words
	GPTZero.me (GPTZero, 2023)	✓	✓	✓	✓		✓	✓	250 chars to 5k chars
	Grover.Allenai (Zellers et al., 2019)	✓					✓	✓	100 chars to 100k chars
	Originality.ai (Originality AI, 2023)			✓	✓		✓		50 words to 300 words
	Scribbr.com (Scribbr, 2023)	✓	✓	✓			✓	✓	25 words to 500 words
	Writer.com (Writer, 2023)					✓	✓	✓	<1.5k chars

original question askers, reflecting community validation of correctness, explanatory clarity, and instructional usefulness. Second, an expert panel consisting of 22 senior engineering professors (each with over ten years of teaching experience across computer science, mechanical engineering, and civil engineering) conducted manual reviews of the candidate questions and answers. The panel focused on identifying submissions that aligned closely with current university curricula, exhibited step-by-step reasoning, and demonstrated pedagogical intent rather than generic or template-like code. Through this process, a total of 42,740 questions and their corresponding accepted answers were retained to form the final DataCode dataset. While we acknowledge that a small amount of AI-assisted content may still exist, the combined temporal constraints, community acceptance mechanism, and expert screening provide a sufficiently reliable approximation of human-written engineering code for the comparative evaluation conducted in this study. We took the answers accepted by the askers in the original posts as human samples and generated AI code solutions using the following prompt:

“You are a programming expert. Please provide a correct and efficient solution to the following programming question. Only return the code and necessary explanations, written clearly and professionally. Question: [Insert Programming Problem].”

In this prompt, the instruction for the model to act as a “programming expert” was not intended to equate engineering students with professional software engineers. Rather, it served as a quality-control constraint to ensure that the generated code was syntactically correct, logically coherent, and executable, matching the expected competency level of upper-division undergraduate and graduate engineering students. The DataCode dataset focuses on advanced coursework and project-level tasks, for which well-prepared students are typically capable of producing complete and correct implementations. Accordingly, this prompt design avoids generating low-quality or partially functional AI outputs and enables a fair and realistic evaluation of detector performance under challenging yet educationally plausible conditions.

By instructing LLMs to act as programming experts, provide correct and efficient code solutions along with code explanations, we ultimately integrated the DataCode dataset, which contains 42,740 pairs of code and explanation samples. It aims to authentically reflect the logical pathways and coding styles of engineering students when solving practical technical problems.

3.2. Selection of AIGC detectors

To comprehensively assess the current ecological landscape of AIGC detection technologies, this study selected 13 mainstream detection tools as research objects based on market share, algorithmic representativeness, and community influence. These tools include 7 open-source detectors based on public algorithms and 6 commercial detectors offering proprietary services.

Open-source detectors primarily rely on statistical features (e.g., perplexity and entropy) or fine-tuned pre-trained language models for discrimination, reflecting the academic community’s exploration of the intrinsic characteristics of AI-generated text. In contrast, commercial detectors are widely adopted in educational institutions and publishing workflows and are typically provided as proprietary systems without publicly disclosed algorithms. For experimental feasibility and reproducibility, all detectors were accessed through their official application programming interfaces (APIs) or publicly available web interfaces. The key characteristics and configurations of each detection tool are summarized in Table 1. It should be noted that the “Detectable Models” column refers to language models explicitly documented or evaluated in official sources, rather than constituting strict functional limitations of the detectors.

3.3. Evaluation metrics

To comprehensively evaluate the performance and reliability of AIGC detection tools across different types of student academic content, this study adopted three quantitative evaluation metrics—Area Under the Curve (AUC), False Positive Rate (FPR), and

False Negative Rate (FNR)—forming a comprehensive evaluation system for the accuracy, fairness, and robustness of detection tools. Given that some detectors only output sentence-level scores, we implemented a Maximum Aggregation Strategy before calculating the metrics: the maximum value of the prediction scores of all sentences in a paragraph was taken as the final judgment score for that paragraph. This strategy reflects the real logic of educational assessment: in academic integrity reviews, even a small number of sentences exhibiting obvious AI-generated characteristics are sufficient to trigger suspicion of the entire paragraph. Based on the aggregated paragraph-level scores, we calculated the following metrics.

3.3.1. AUC score

AUC (Area Under the ROC Curve) measures the overall discriminative ability of detection tools across all possible classification thresholds. It represents the probability that a detector will classify a randomly selected AI-generated sample as “more likely to be AI than a randomly selected human-written sample”. In the evaluation system of this study, we defined the Positive Class as LLM-generated content and the Negative Class as human-written student assignments. A higher AUC value indicates that the detector possesses a stronger ability to distinguish between AI-generated texts and authentic human academic writing across diverse datasets such as StuTask, StuThesis, and DataCode.

Specifically, the calculation of the AUC value is based on the prediction scores of the detector across multiple classification thresholds. For each threshold, we calculate the True Positive Rate (TPR) and False Positive Rate (FPR) respectively, where $TPR = 1 - FNR$. Subsequently, a Receiver Operating Characteristic (ROC) curve is plotted with FPR on the x -axis and TPR on the y -axis, and the Trapezoidal Rule is used to integrate the curve to derive the final value. This method aims to evaluate the overall discriminative ability of the detector, thereby avoiding the limitations associated with relying on a single decision threshold.

3.3.2. FPR

The False Positive Rate (FPR) refers to the proportion of human-written content that is incorrectly identified as AI-generated. Its calculation formula is as follows:

$$FPR = \frac{FP}{FP + TN} \quad (1)$$

where FP (False Positives) denotes the number of human-written samples incorrectly classified as AI-generated, and TN (True Negatives) represents the number of human-written samples correctly classified. In an educational context, FPR is a key metric for measuring assessment fairness. A high FPR means students' independently completed assignments are highly likely to be falsely flagged as cheating, which can lead to unjust academic penalties and serious ethical issues. Therefore, in academic integrity assessments, maintaining an extremely low FPR is often more important than pursuing a high detection rate.

3.3.3. FNR

The False Negative Rate (FNR) measures the proportion of AI-generated content that is incorrectly classified as human-written. Its calculation formula is as follows:

$$FNR = \frac{FN}{FN + TP} \quad (2)$$

where FN (False Negatives) refers to the number of AI-generated samples that are missed, and TP (True Positives) denotes the number of AI-generated samples that are correctly identified. The FNR directly reflects the effectiveness of detection tools. A high FNR indicates that the detector fails to identify most AI-generated texts, making it unable to effectively uphold academic integrity. In our analysis, a low FNR means that even if students attempt to conceal traces of AI generation, the model still maintains high detection sensitivity.

4. Results

4.1. Effectiveness of existing detectors (RQ1)

To address the research question “How effective are existing detection tools in identifying LLM-generated content”, we conducted comprehensive tests on 13 mainstream detectors using the three constructed datasets (StuTask, StuThesis, and DataCode). The tested tools include 7 open-source models and 6 commercial closed-source tools. Table 2 presents the performance of each detector in terms of Area Under the Curve (AUC), False Positive Rate (FPR), and False Negative Rate (FNR) across different datasets.

In terms of the average AUC, the detectors exhibit significant performance variations across different assignment genres. All tools achieve the best performance on StuThesis, with an average AUC of 71.48%; followed by StuTask (short assignments) at 57.85%; and the worst performance on DataCode (code assignments), where the average AUC is only 54.71%, hovering around the level of random guessing (50%). This trend indicates that the discriminative ability of detectors is highly dependent on the length and richness of the text. StuThesis, with its longer average length (Human: 817 words/AI: 984 words), provides more statistical features for model analysis, enabling tools such as Scribbr (96.77%) and GPTZero (95.01%) to achieve near-perfect detection on this type of task. In contrast, due to the short length and sparse features of StuTask texts, the performance of many detectors degrades to near-random guessing—for instance, GPT2-Detector has an AUC of only 32.75%.

In the comparison between open-source and commercial tools, we observed an interesting “divergence” phenomenon. Overall, commercial tools generally outperform older-generation open-source models in text detection. Among open-source detectors, while early models (e.g., GPT2-Detector, ArguGPT) exhibit poor performance, the latest Fast-Detect GPT demonstrates excellent results,

Table 2
Performance of 13 detectors across three datasets.

Detector	StuTask			StuThesis			DataCode		
	AUC	FPR	FNR	AUC	FPR	FNR	AUC	FPR	FNR
GPT2_Detector	32.75	9.95	80.32	77.03	32.17	29.84	57.68	32.05	56.77
RoBERTa-QA	54.05	29.03	58.47	42.02	16.05	68.41	32.49	90.07	21.25
ArguGPT	29.42	71.03	32.96	95.08	8.52	24.06	63.73	18.06	39.41
RADAR	40.62	46.07	48.20	36.82	93.02	10.74	52.56	83.06	21.23
MAGE	81.49	29.05	20.21	92.05	14.92	21.26	67.82	22.49	56.02
MultiScale	39.06	61.37	52.08	31.58	42.75	69.14	39.05	64.93	21.05
Fast-DetectGPT	90.03	14.05	18.62	92.26	4.06	9.77	79.47	72.91	27.08
Average	52.49	37.22	44.41	66.69	30.21	33.32	56.11	54.80	34.69
Crossplag.com	72.82	39.43	11.05	68.52	52.85	19.07	56.24	45.83	10.64
GPTZero.me	74.00	23.16	30.94	95.01	9.74	8.06	45.79	42.06	73.97
Grover	50.28	84.90	10.02	50.00	100.00	0.00	53.06	100.00	0.00
Originality.ai	58.42	32.09	20.15	88.52	21.86	3.47	58.73	12.74	49.61
Scribbr.com	49.26	40.92	49.05	96.77	2.48	3.96	76.52	17.05	32.18
Writer.com	79.88	20.04	16.83	63.59	4.38	42.05	28.04	8.04	78.42
Average	64.11	40.09	23.01	77.07	31.89	12.77	53.06	37.62	40.80
Overall average	57.85	38.55	34.53	71.48	30.98	23.83	54.71	46.87	37.51

maintaining extremely high accuracy across all three datasets: StuTask (90.03%), StuThesis (92.26%), and DataCode (79.47%), with its False Positive Rate (FPR) controlled within a reasonable range. This indicates that the zero-shot detection method based on conditional probability curvature possesses strong generalization ability. Regarding closed-source tools, Scribbr and GPTZero achieve remarkably low False Negative Rates (FNR) of 3.96% and 8.06% respectively on StuThesis, meaning they can identify the vast majority of AI-generated thesis paragraphs. Despite the overall high average scores of closed-source tools, their average FPR on the StuTask dataset reaches as high as 40.09%, even exceeding the 37.22% of open-source tools. For instance, Writer.com's AUC on DataCode drops to 28.04%, rendering it almost completely ineffective. This implies that in real-world assignment evaluations, using these “black-box” tools is highly likely to result in honest students being falsely accused, which constitutes a dangerous signal in educational assessment. This finding challenges the stereotype that “paid commercial tools are necessarily superior to open-source academic models” and demonstrates the superiority of the zero-shot detection method based on conditional probability curvature.

One of the most notable findings of this study is that existing tools collectively suffered a near total setback on the DataCode dataset. Except for Scribbr (76.52%) and Fast-DetectGPT (79.47%), the AUC of the vast majority of detectors falls below 60%, hovering around the level of random guessing. As a leading tool in text detection, GPTZero achieves an AUC of only 45.79% in code detection, with a remarkably high False Negative Rate (FNR) of 73.97%. This means that over 70% of AI-generated code can successfully evade GPTZero's scrutiny. The performance of RoBERTa-QA (32.49%) and Writer.com (28.04%) even falls below the random level (AUC < 50%), which strongly demonstrates that there is an inherent difference between the rigorous logical structure of programming languages and the statistical patterns of natural language. Directly transferring existing text detectors to audit engineering code assignments is therefore highly unreliable.

Based on the overall test results, there is currently no “all-purpose” AIGC detector available. While Fast-DetectGPT and Scribbr perform excellently in specific tasks, most tools suffer from severe accuracy deficiencies when faced with short assignments and code-related tasks. Particularly for engineering disciplines, relying on existing general-purpose detectors for academic integrity reviews carries an extremely high risk of misjudgment. This result strongly calls for caution among the education community when using such tools and advocates for the adoption of differentiated assessment strategies tailored to different disciplines.

 **Answer to RQ1:** The findings indicate that the performance of AIGC detectors varies significantly across different assignment genres, with commercial tools generally outperforming open-source models, but existing detectors still struggling with short assignments and code-related tasks, highlighting the need for discipline-specific detection strategies in academic integrity assessments.

4.2. Factors affecting detection performance (RQ2)

To further explore which intrinsic features of generated content interfere with detectors' judgments, this section examines the specific impacts of Cognitive Complexity, Rhetorical Function, and Structural Length on detection performance through controlled variable analysis. Tables 3 to 6 detail the modified performance fluctuations of the detectors' AUC across different subsets of StuTask, StuThesis, and DataCode. The values in parentheses indicate the relative AUC change compared to the overall baseline performance in Table 2, where “+” indicates an improvement compared to the results in RQ1, and “−” indicates a decline.

4.2.1. Impact of task difficulty: The cognitive complexity hypothesis

In the StuTask dataset, we divided the questions into three levels based on students' historical score rates: Easy (>80%), Medium (60%–80%), and Hard (<60%), which served as a proxy variable for the cognitive complexity of the tasks. The experimental results are presented in Table 3.

Table 3

The AUC performance of different detectors on the StuTask dataset.

Detector	Easy	Medium	Hard
GPT2_Detector	34.72 (+1.97)	33.96 (+1.21)	29.57 (−3.18)
RoBERTa-QA	65.37 (+11.32)	56.49 (+2.44)	40.29 (−13.76)
ArguGPT	23.49 (−5.93)	25.08 (−4.34)	39.69 (+10.27)
RADAR	42.93 (+2.31)	39.61 (−1.01)	39.32 (−1.30)
MAGE	86.97 (+5.48)	84.02 (+2.53)	73.48 (−8.01)
MultiScale	45.02 (+5.96)	44.16 (+5.10)	28.00 (−11.06)
Fast-DetectGPT	90.68 (+0.65)	89.75 (−0.28)	89.66 (−0.37)
Average	55.60	53.30	48.57
Crossplag.com	70.16 (−2.66)	70.72 (−2.10)	77.58 (+4.76)
GPTZero.me	75.93 (+1.93)	75.01 (+1.01)	71.06 (−2.94)
Grover	54.09 (+3.81)	50.16 (−0.12)	46.59 (−3.69)
Originality.ai	58.04 (−0.38)	58.73 (+0.31)	58.49 (+0.07)
Scribbr.com	59.58 (+10.32)	54.19 (+4.93)	34.01 (−15.25)
Writer.com	79.42 (−0.46)	80.07 (+0.19)	80.15 (+0.27)
Average	66.20	64.81	61.31
Overall average	60.49	58.61	54.45

Table 4

The AUC performance of different detectors on the StuThesis dataset.

Detector	Introductory	Methodological	Reflective
GPT2_Detector	75.27 (−1.76)	82.84 (+5.81)	72.98 (−4.05)
RoBERTa-QA	47.32 (+5.30)	57.42 (+15.40)	21.32 (−20.70)
ArguGPT	95.82 (+0.74)	96.75 (+1.67)	92.67 (−2.41)
RADAR	34.71 (−2.11)	36.92 (+0.10)	38.83 (+2.01)
MAGE	92.17 (+0.12)	94.58 (+2.53)	89.42 (−2.65)
MultiScale	29.12 (−2.48)	36.93 (+5.35)	28.71 (−2.87)
Fast-DetectGPT	91.73 (−0.53)	90.86 (−1.40)	94.19 (+1.93)
Average	66.59	70.90	62.59
Crossplag.com	68.26 (−0.26)	68.19 (−0.33)	69.11 (+0.59)
GPTZero.me	94.06 (−0.95)	95.93 (+0.92)	95.04 (+0.03)
Grover	49.99 (−0.01)	49.99 (−0.01)	50.02 (+0.02)
Originality.ai	88.39 (−0.13)	86.03 (−2.49)	91.14 (+2.62)
Scribbr.com	98.02 (+1.25)	97.16 (+0.39)	95.13 (−1.64)
Writer.com	58.36 (−5.23)	69.14 (+5.55)	63.27 (−0.32)
Average	76.18	77.74	77.29
Overall average	71.02	74.06	69.37

The experimental results indicate a negative correlation between detector performance and task difficulty: the higher the task difficulty, the lower the detector effectiveness. Among all tested models, the vast majority of tools achieve the best performance when handling “Easy” tasks, while their performance is poorest when confronted with “Hard” tasks. Taking Scribbr.com as an example, its AUC reaches a moderate 59.58% for Easy questions but plummets to 34.01% for Hard questions. Similarly, RoBERTa-QA’s performance drops from 65.37% on Easy tasks to 40.29% on Hard tasks. In contrast, Fast-DetectGPT demonstrates exceptional anti-interference capability, maintaining a high accuracy of approximately 90% across all three difficulty levels and remaining almost unaffected by task complexity. Hard tasks typically require more rigorous logical reasoning, longer explanation chains, and precise use of technical terminology. For such tasks, high-quality LLM-generated answers are highly convergent with the in-depth thinking of top students in terms of linguistic features, thereby blurring the classification boundary and making it difficult for detectors to effectively distinguish between them through superficial statistical features.

4.2.2. Impact of writing section: Rhetorical function and formulaic language

Each thesis was categorized into three types based on its rhetorical and structural position in academic papers: Introductory Sections (Introduction, Literature Review)—these sections typically precede the Methodology and Results, containing conceptual overviews, definitions, and a comprehensive analysis of existing research; Methodological Sections (Methodology, Experimental Results)—these sections emphasize the precision of procedural descriptions and analyses, characterized by the use of formulaic and domain-specific expressions; and Reflective Sections (Discussion, Conclusion)—these sections follow the Results, focusing on interpretation, implications, and summary. As shown in Table 4, in the StuThesis dataset, we compared the detection performance across Introductory, Methodological, and Reflective chapters and found that rhetorical function significantly moderates detection accuracy.

The experimental results indicate that different chapters of academic papers pose distinctly different challenges to detectors due to their unique discourse functions. Overall, Methodological chapters focusing on procedural descriptions are the easiest to detect,

Table 5
Attribution-based cue utilization in StuThesis (Methodology vs. Discussion).

Section	Academic/Template	Structural	Numeric/Citation	Other lexical
Methodology	0.42	0.27	0.18	0.13
Discussion	0.26	0.16	0.06	0.51
Δ (Meth – Disc)	+0.16	+0.11	+0.12	−0.38

Notes. Values denote the proportion of total absolute attribution mass assigned to each cue category, computed using Integrated Gradients and averaged over 300 randomly sampled thesis paragraphs per section.

with most detectors achieving peak AUC values here. For instance, RoBERTa-QA’s performance in Methodological sections (57.42%) is significantly superior to that in Introductory (47.32%) and Discussion (21.32%) sections. In contrast, detection performance on Reflective chapters exhibits substantial variability: Fast-DetectGPT achieves the highest overall AUC of 94.19% in this section, while RoBERTa-QA plummets to the lowest overall value of 21.32%, reflecting differences in how various technical routes handle subjective text.

Methodological writing features a high degree of formulaicity and fixed academic conventions. When generating such text, LLMs often fall into specific probabilistic patterns, which are easily captured by statistical models. In contrast, Discussion sections—characterized by strong subjectivity and a focus on meaning-making—contain a wealth of authors’ personal viewpoints and complex arguments. These text features, marked by high flexibility, are a fatal interference for models relying on specific training data (e.g., RoBERTa-QA), but they instead provide richer discriminative signals for zero-shot models based on probability curvature (e.g., Fast-DetectGPT).

To further substantiate the above interpretation and move beyond purely performance-based observations, we conducted an attribution-based interpretability analysis focusing specifically on Methodology and Discussion sections in the StuThesis dataset. Concretely, we applied Integrated Gradients as a post-hoc attribution method to analyze the decision signals of representative detectors at the token level. For each section type, we randomly sampled 300 thesis paragraphs and computed token attributions with respect to the predicted class. Absolute attribution scores were then aggregated and normalized to quantify the relative contribution of different cue categories. The aggregated attribution distributions for the two section types are summarized in Table 5. Results were averaged across all sampled paragraphs within each section, and the observed trends were highly consistent across samples, indicating that the patterns reported below are not driven by isolated examples.

The analysis reveals a pronounced asymmetry in cue utilization between Methodology and Discussion sections. In Methodology paragraphs, detector decisions are dominated by academic template cues, structural markers, and numeric or citation-related signals. Typical high-attribution tokens include formulaic expressions such as “*in this study*”, “*we propose*”, and “*results indicate*”; discourse scaffolding markers such as “*first*”, “*therefore*”, and “*however*”; as well as numeric and symbolic elements including digits, brackets, and references to “*Table 3*” or “*Fig. 2*”. These elements reflect the rigid and highly standardized nature of procedural academic writing, where both human-written and AI-generated texts tend to converge on similar lexical and structural patterns.

In contrast, Discussion sections exhibit a fundamentally different attribution profile. Here, a substantial proportion of attribution mass shifts away from structural and template-based cues toward content-specific lexical items that encode interpretation, stance, and evaluation. High-attribution tokens frequently include evaluative phrases such as “*raises concerns*”, “*suggests that*”, “*may undermine*”, or “*is indicative of*”, as well as domain-specific concepts whose usage depends strongly on authorial intent and argumentative context. Unlike Methodology writing, these expressions are less constrained by fixed conventions and display greater stylistic and semantic variability, reducing the effectiveness of detectors that depend on recurring surface patterns.

This contrast provides mechanistic evidence for the performance gap observed in Table 4. Detectors achieve higher accuracy on Methodology sections not because these sections are inherently more “AI-like”, but because their formulaicity amplifies statistical regularities that detection models are designed to capture. When such regularities are attenuated—as in Discussion sections, where meaning construction and subjective judgment dominate—the same detectors lose reliable decision signals. Importantly, this finding reinforces our central argument: current AIGC detectors primarily operate on shallow structural cues rather than deep semantic understanding, which fundamentally limits their reliability in reflective and interpretive academic writing.

4.2.3. Impact of code length: The structural complexity dilemma

For the DataCode dataset, we investigated how the code length required to solve a problem affects the detectability of AI-generated answers. We first analyzed the token counts of all accepted human-written answers on Stack Overflow, where a “token” refers to a syntactic unit recognized by the tokenizer used in LLM input processing—such as keywords, variables, operators, or punctuation marks. The 33rd and 67th percentiles of the token length distribution were approximately 72 tokens and 197 tokens, respectively. We therefore divided the code length into three ranges: Short (0–72 tokens), Medium (72–186 tokens), and Long (186 tokens and above). Based on the number of lines or tokens of human-written code, we split the dataset into three subsets. The results in Table 6 show that the performance of nearly all tools degrades significantly as code length increases.

In code detection tasks, code length (Token Count) has been proven to be the most critical variable determining detection success, with a strict monotonically decreasing relationship between the two. Nearly all detection tools achieve optimal performance on “Short Code” (<72 tokens) but perform poorest on “Long Code” (>186 tokens). Scribbr.com’s AUC reaches as high as 85.49% for short code but drops significantly to 64.23% for long code; Writer.com even plummets from 41.06% to 19.57%, becoming completely ineffective. Even the most robust Fast-DetectGPT experiences a slight performance decline as length increases (from 80.94% to

Table 6
The AUC performance of different detectors on the DataCode dataset.

Detector	Short	Medium	Long
GPT2_Detector	60.52 (+2.84)	56.19 (−1.49)	56.33 (−1.35)
RoBERTa-QA	44.17 (+11.68)	36.02 (+3.53)	17.28 (−15.21)
ArguGPT	68.50 (+4.77)	65.52 (+1.79)	57.17 (−6.56)
RADAR	54.05 (+1.49)	52.16 (−0.40)	51.47 (−1.09)
MAGE	71.04 (+3.22)	67.19 (−0.63)	65.23 (−2.59)
MultiScale	52.48 (+13.43)	34.29 (−4.76)	30.38 (−8.67)
Fast-DetectGPT	80.94 (+1.47)	79.81 (+0.34)	77.66 (−1.81)
Average	61.67	55.88	50.79
Crossplag.com	59.26 (+3.02)	57.43 (+1.19)	52.03 (−4.21)
GPTZero.me	46.69 (+0.90)	45.02 (−0.77)	45.66 (−0.13)
Grover	56.94 (+3.88)	52.79 (−0.27)	49.45 (−3.61)
Originality.ai	59.62 (+0.89)	58.14 (−0.59)	58.43 (−0.30)
Scribbr.com	85.49 (+8.97)	79.84 (+3.32)	64.23 (−12.29)
Writer.com	41.06 (+13.02)	23.49 (−4.55)	19.57 (−8.47)
Average	58.18	52.79	48.23
Overall average	60.06	54.45	49.61

Table 7
Attribution-based cue utilization across different code lengths in DataCode.

Length	Symbol/Format	Keyword frequency	Repetition patterns	Other tokens
Short	0.42	0.30	0.17	0.11
Medium	0.29	0.26	0.27	0.18
Long	0.21	0.13	0.19	0.47
Δ (Short – Long)	+0.21	+0.17	−0.02	−0.36

Notes. Values denote the proportion of total absolute attribution mass assigned to each cue category, computed using Integrated Gradients and averaged over 300 randomly sampled code instances per length group.

77.66%). In detail, short code typically involves only simple syntax exercises or standard algorithm implementations, where LLM generation patterns are relatively uniform and easy to identify. However, as code length increases, the involved modular design, logical nesting, and exception handling become extremely complex. The high degree of freedom in long code provides LLMs with more “humanization” space, making the complex logical structures they generate indistinguishable from those written by human experts—thus leading to the poor performance of detectors when faced with long engineering code.

To further explain the pronounced performance degradation observed on long code segments, we conducted an attribution-based interpretability analysis on the DataCode dataset, following the same analytical framework used for StuThesis. Using Integrated Gradients, we analyzed token-level attribution patterns for representative detectors and randomly sampled 300 code instances per length category. Attribution scores were computed with respect to the predicted class, aggregated in absolute value, and normalized within each sample. Results were then averaged across all samples in each length group. As summarized in Table 7, detector decisions for Short Code are dominated by surface-level syntactic and lexical cues. High-attribution tokens frequently include programming symbols and formatting markers such as `{`, `}`, `;`, `==`, `->`, as well as high-frequency keywords like `if`, `for`, `return`, and basic data types (e.g., `int`, `float`). These elements are characteristic of short, self-contained code snippets that implement standard algorithms or textbook-style exercises, where both human-written and LLM-generated solutions tend to follow highly regularized patterns.

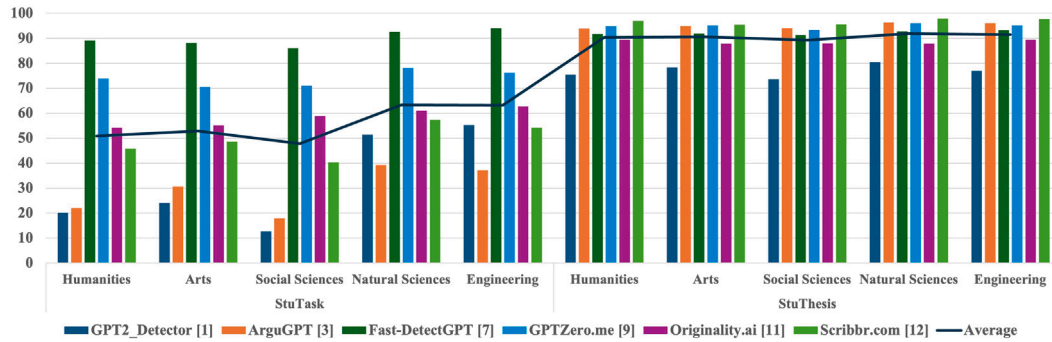
For Medium Code, attribution mass begins to diffuse away from purely symbolic cues toward repeated variable names, function headers, and control-flow scaffolding. While syntactic markers remain salient, detectors increasingly attend to patterns such as repeated loop constructs, function signatures, and recurring identifier names, reflecting an implicit reliance on structural repetition as a proxy for detectability. This transition coincides with a noticeable decline in detection accuracy, as shown in Table 6, suggesting that reliance on such mid-level structural regularities is already less stable.

In Long Code, detector attribution patterns undergo a more fundamental shift. No single category of surface cues consistently dominates the decision process; instead, attribution mass is fragmented across diverse tokens, including variable identifiers, nested control structures, and exception-handling logic. Notably, many of these tokens carry little intrinsic statistical distinctiveness, as long code solutions—whether generated by LLMs or written by human experts—exhibit substantial diversity in naming conventions, modular decomposition, and logical organization. As a result, detectors that primarily rely on symbol frequency, repetition, or formatting cues lose reliable signals, leading to the near-random performance observed for most tools on long code segments.

Table 8

The amount of data for each discipline under the two datasets.

Discipline	StuTask	StuThesis
Humanities	5082	12 648
Arts	4615	10 336
Social Sciences	5459	17 472
Natural Sciences	5741	18 930
Engineering	5140	15 195
Total	26 037	74 581

**Fig. 2.** The AUC performance of different detectors on both datasets.

Answer to RQ2: The controlled analyses across various datasets reveal three critical factors that systematically impair detector performance: the cognitive complexity of generated content (difficult tasks), specific rhetorical features (reflective writing), and structural intricacies (long code). These findings suggest that current AIGC detection tools perform effectively on simpler, standardized tasks but exhibit a significant decline in reliability when dealing with outputs requiring advanced reasoning and expert-level problem-solving.

4.3. Disciplinary effects on generated content (RQ3)

To further explore how disciplinary linguistic patterns influence the detectability of AI-generated content, we divided the StuTask and StuThesis datasets into five major academic disciplines: Humanities, Arts, Social Sciences, Natural Sciences, and Engineering, with the distribution of samples across disciplines shown in Table 8.

In this section, we selected six representative detectors for horizontal comparison, including three open-source models: Fast-DetectGPT, GPT2-Detector, and ArguGPT; and three commercial platforms: GPTZero.me, Originality.ai, and Scribbr.com. Fast-DetectGPT was chosen because it consistently demonstrated the most robust and stable performance across all primary datasets (RQ1), making it a reliable benchmark for cross-disciplinary comparison. Second, the remaining five detectors were included because they exhibited significant performance inconsistency across different genres—performing poorly on short coursework (StuTask) but achieving high accuracy on long academic essays (StuThesis). This variability presents a crucial research opportunity: detectors that are sensitive to text length or genre may also respond differently to the unique linguistic habits, vocabulary density, and structural norms of different academic disciplines. By incorporating both the top-performing benchmark and those with unstable performance, our design enables a more nuanced understanding of how domain-specific linguistic patterns influence AIGC detection.

Across the five disciplines, we compared the AUC scores of all six detectors. This experimental setup enables a detailed analysis of how domain-specific linguistic patterns (e.g., disciplinary vocabulary density, conceptual abstraction, and structural regularity) influence detection accuracy. Disciplines such as Engineering and Natural Sciences tend to exhibit more formulaic expressions and technical terminology, which may increase the identification difficulty for detectors primarily trained on general English corpora. In contrast, Humanities and Arts writing typically demonstrates stylistic variation and subjectivity, which may render AI-generated text either easier or harder to identify—depending on the underlying linguistic sensitivity of the detector. Thus, this design provides a comprehensive foundation for understanding how disciplinary contexts shape the behavior and fairness of AIGC detectors in educational assessment.

From the disciplinary perspective in Fig. 2, detectors exhibit a significant “arts-sciences divide” characteristic, with STEM fields emerging as the biggest blind spot for detectors. Data shows that the accuracy of most commercial detectors declines sharply in Engineering and Natural Sciences, yielding their worst performance. Taking GPTZero.me and Originality.ai as examples, these two closed-source tools—which perform excellently in general writing—experience drastic fluctuations in AUC when confronted with engineering texts: not only does their detection rate decrease, but their false positive rate also becomes unstable. In contrast, the open-source model Fast-DetectGPT demonstrates the best robustness, maintaining high accuracy (AUC > 85%) in STEM fields comparable to that in humanities and social sciences. Scientific and engineering writing tends to feature high formulaicity and

dense technical terminology in logical reasoning and expression. Within the academic norms of Natural Sciences and Engineering, both human-written and AI-generated texts tend to adopt standardized definitions, objective passive voice, and fixed reasoning steps. This writing style—characterized by the abandonment of personal style and the exposition of objective logic and content—naturally reduces the text's perplexity and burstiness. For commercial detectors primarily relying on these statistical features, the rigorous structural characteristics of scientific and engineering texts are prone to causing algorithmic confusion, making it difficult to distinguish between rigid AI generation and rigorous human expression, thereby leading to detection failure.

In contrast to science and engineering disciplines, the Humanities and Arts are characterized by rich emotional expression and personal stylistic features, which provide more discriminative signals for detectors but also introduce new risks. Overall, early models (e.g., GPT2-Detector) perform best on humanities texts, with their detection success rates significantly higher than those in science and engineering. However, for arts texts, the False Positive Rate (FPR) of some commercial tools shows an upward trend, indicating that creative expressions are sometimes incorrectly flagged as machine-generated. On one hand, humanities writing typically incorporates rich emotional overtones, metaphors, and complex rhetorical devices. Since current LLMs still lack sophistication in mimicking humans' deep emotional resonance and unique personal styles, these differences enable classifiers based on simple statistical features to more easily identify traces of AI generation. On the other hand, in artistic creation, students may employ highly innovative unconventional expressions or non-standard grammar. For black-box models trained on standard academic English, these human innovative expressions are highly prone to being recognized as statistical outliers by algorithms, thereby increasing the FPR and raising concerns about assessment fairness.

In the horizontal comparison of interdisciplinary adaptability, detectors with different technical routes have demonstrated distinctly different generalization capabilities. Fast-DetectGPT, the top-performing detector, exhibits minimal performance fluctuations across all five disciplines, proving the universality of the zero-shot method. In contrast, commercial tools such as Originality.ai and GPTZero.me show obvious “disciplinary bias”—significant performance gaps between humanities/social sciences and science/engineering, indicating high instability. Domain bias remains a prevalent issue in detector training. Closed-source detectors are typically trained on massive amounts of general English text, so they perform reasonably well when handling humanities and social sciences assignments with similar styles. However, when confronted with highly specialized science and engineering texts featuring unique linguistic patterns, the prior knowledge of their pre-trained models becomes mismatched, leading to degraded generalization ability. In contrast, Fast-DetectGPT directly leverages the inherent probability distribution of LLMs for judgment without relying on domain-specific external training data, thus enabling it to cross disciplinary boundaries and achieve more stable detection.

 **Answer to RQ3:** The findings reveal that AIGC detectors exhibit significant disciplinary biases, with commercial tools performing poorly in STEM fields, while the open-source model Fast-DetectGPT demonstrates superior interdisciplinary adaptability, maintaining consistent accuracy across diverse academic disciplines.

4.4. Detector robustness against content modifications (RQ4)

In real-world educational settings, students rarely submit unmodified raw LLM outputs. Instead, they typically make minor revisions—polishing the writing style, adjusting wording, or adding subtle “noise”—to make the text appear more human-written and reduce the probability of AI detection. Similarly, human-written content may also undergo partial modifications using AI-powered editing tools, which can inadvertently alter how detectors classify the text. Therefore, it is crucial to test whether AIGC detectors remain reliable when the surface form of the text is slightly altered. In this experiment, we investigated the robustness of detectors after applying minor, human-like edits to both LLM-generated and human-written content.

To ground our robustness study in real student behaviors, we conducted semi-structured interviews with 25 undergraduate and graduate students who frequently use AI tools for text polishing and generation. We asked them to describe the specific strategies they employ to “lower AI scores” or make AI-generated text appear more human-written. Based on their responses, we summarized the five most frequently mentioned strategies and denote these modification strategies as **RSP** (Remove Spaces between words or around punctuation), **NIN** (Noise Insertion with Numbers/Letters), **TSM** (Tense Shift Modification), **SYN** (Synonym and Colloquial Replacement), and **SRS** (Sentence Restructuring and Splitting), respectively.

- **RSP.** Removing spaces between coordinated or adjacent words (e.g., compressing “pros and cons” into “pros&cons” or reducing spaces around punctuation), which slightly alters tokenization patterns without changing meaning.
- **NIN.** Inserting irrelevant digits or letters at non-disruptive positions (e.g., adding small numbers, single letters, or short sequences that do not affect comprehension), thereby introducing noise into the text structure.
- **TSM.** Modifying the tense of selected sentences (e.g., switching from past to present or to present perfect), while keeping the underlying semantics intact.
- **SYN.** Replacing key words or phrases with simpler or more colloquial synonyms, making the text less formal and closer to typical student writing.
- **SRS.** Restructuring sentences by splitting, merging, or lightly reordering clauses, such as breaking a long sentence into two shorter ones or adding discourse markers (e.g., “actually”, “in fact”, “to be honest”) to mimic natural human flow.

Table 9
Robustness analysis results under five text modification strategies.

Strategy	AI-generated			Human		
	Num	Detector	AI_{rate}	Num	Detector	H_{rate}
Total	10 485	–	–	3371	–	–
RSP	4873	GPT2_Detector	0.45	983	GPT2_Detector	0.07
		ArguGPT	0.51		ArguGPT	0.04
		Fast-DetectGPT	0.54		Fast-DetectGPT	0.04
		GPTZero.me	0.49		GPTZero.me	0.06
		Originality.ai	0.52		Originality.ai	0.02
		Scribbr.com	0.40		Scribbr.com	0.04
NIN	3209	GPT2_Detector	0.32	805	GPT2_Detector	0.01
		ArguGPT	0.25		ArguGPT	0.00
		Fast-DetectGPT	0.27		Fast-DetectGPT	0.01
		GPTZero.me	0.44		GPTZero.me	0.00
		Originality.ai	0.21		Originality.ai	0.00
		Scribbr.com	0.36		Scribbr.com	0.00
TSM	6050	GPT2_Detector	0.12	1417	GPT2_Detector	0.02
		ArguGPT	0.16		ArguGPT	0.01
		Fast-DetectGPT	0.05		Fast-DetectGPT	0.01
		GPTZero.me	0.08		GPTZero.me	0.00
		Originality.ai	0.19		Originality.ai	0.00
		Scribbr.com	0.07		Scribbr.com	0.00
SYN	5194	GPT2_Detector	0.42	758	GPT2_Detector	0.00
		ArguGPT	0.48		ArguGPT	0.00
		Fast-DetectGPT	0.59		Fast-DetectGPT	0.00
		GPTZero.me	0.54		GPTZero.me	0.00
		Originality.ai	0.37		Originality.ai	0.00
		Scribbr.com	0.45		Scribbr.com	0.00
SRS	3065	GPT2_Detector	0.26	842	GPT2_Detector	0.00
		ArguGPT	0.18		ArguGPT	0.00
		Fast-DetectGPT	0.28		Fast-DetectGPT	0.01
		GPTZero.me	0.21		GPTZero.me	0.00
		Originality.ai	0.13		Originality.ai	0.00
		Scribbr.com	0.11		Scribbr.com	0.00
Hybrid	5746	GPT2_Detector	0.68	1295	GPT2_Detector	0.00
		ArguGPT	0.74		ArguGPT	0.09
		Fast-DetectGPT	0.88		Fast-DetectGPT	0.13
		GPTZero.me	0.79		GPTZero.me	0.09
		Originality.ai	0.71		Originality.ai	0.05
		Scribbr.com	0.64		Scribbr.com	0.07

For each strategy, we applied the corresponding transformation to a subset of both LLM-generated and human-written texts to evaluate how well the detectors withstand behaviorally plausible edits. To operationalize this process, we first randomly sampled instances from our datasets that were suitable for each modification type, and then applied exactly one of the five strategies to each selected sample. In addition, we designed a Hybrid strategy, in which each sentence within a sample was randomly assigned one of the five modification operations; this random mutation process was repeated five times per sample to simulate diverse and realistic editing behaviors.

To quantify robustness, we measured how many originally detectable samples became undetectable after modification. Specifically, AI_{rate} denotes the proportion of the 10,485 originally detectable GPT-3.5-generated samples that could no longer be identified as AI-generated after mutation. Similarly, H_{rate} represents the proportion of the 3371 originally detectable human-written samples that were no longer recognized as human-written after mutation. The Hybrid condition follows the same definition, but the mutation is applied by randomly selecting one operation from the five defined strategies for each sentence, repeated across five independently mutated versions of the same sample. We utilized the same six representative detectors identified in RQ3 to validate the robustness results, which are presented in Table 9. This consistency allows us to observe whether the “disciplinary blind spots” identified previously correlate with a higher vulnerability to intentional human modifications, thereby providing a holistic view of detector reliability.

The results show that detectors are vulnerable when confronted with single-strategy attacks, with “Synonym Replacement (SYN)” proven to be the most effective single evasion technique. Even the simplest lexical adjustments can drastically degrade detectors’ identification capabilities. Fast-DetectGPT performed the worst against SYN attacks, with an AI Rate as high as 59%; more than half of the AI-generated texts successfully bypassed this model’s scrutiny merely by replacing near-synonyms. Commercial tools were equally unaffected—GPTZero.me’s evasion rate also reached 54%. Additionally, simple formatting interference, such as Space Adjustment (RSP), had a significant impact on older-generation models: for instance, GPT2-Detector’s evasion rate surged to 45% under RSP attacks. Existing detection algorithms are highly reliant on “lexical distribution”. Current statistical detectors primarily

operate by calculating the probability of specific lexical sequence occurrences. Synonym Replacement (SYN) directly alters text's lexical choices, breaking the inherent high-frequency word combination patterns of LLM-generated content and thereby reducing text predictability, making it more statistically similar to human writing. While strategies like RSP and NIN do not change semantic meaning, they interfere with the tokenizer's segmentation logic, causing variations in the token sequence input to the model. This disruption further undermines subsequent probability calculations, resulting in detection failure.

Combined strategies simulate the process of students repeatedly revising and polishing texts in real writing. Such multi-dimensional perturbations produce a Cumulative Effect: Sentence Restructuring (SRS) disrupts sentence structures, Tense Shift Modification (TSM) alters verb forms, and Synonym Replacement (SYN) dilutes high-frequency word density. This series of operations completely erases the “observable traces” remaining in LLM-generated texts. When the statistical features of texts are artificially smoothed out, detectors lose the mathematical basis for judgment, thereby degenerating into random guessing—or even misclassifying polished human-written texts due to excessive sensitivity.

When multiple modification strategies are combined, detectors are essentially unable to resist or identify AI-generated content—this is the most severe performance failure point observed in this study. After launching attacks with a combination of various modification strategies, Fast-DetectGPT, the most robust detector, became the one with the highest evasion rate, whose AI evasion rate soared to 88%. In other words, nearly 9 out of every 10 deeply polished AI assignments successfully “evaded detection”. The performance of closed-source tools is equally worrying: the evasion rates of GPTZero.me and Originality.ai reached 79% and 71%, respectively.

Crucially, this study finds a significant “asymmetry” in how detectors respond to these modifications. While the detectability of AI-generated content collapses, the majority of detectors remain remarkably robust in identifying human-authored text. Except for Fast-DetectGPT, which saw its H-rate rise to 13%, most tools maintain a consistently low and acceptable false positive rate, typically ranging from 0% to 4%. This performance gap provides a strong technical incentive for students to adopt AI-assisted polishing: they can effectively “launder” AI-generated segments to bypass scrutiny while facing minimal risk of their original, independently written work being misclassified. For students, this represents a “high-reward, low-risk” strategy—AI-generated parts become effectively stealthy, while the human-written components remain safely categorized as authentic. Consequently, the widespread adoption of AI tools in higher education is driven not only by their generative efficiency but also by the perceived safety provided by the detectors' own stability in processing human-written content, which inadvertently rewards human-AI collaborative evasion.

 **Answer to RQ4:** The findings confirm that current AIGC detection tools lack robustness against adversarial attacks. The probability statistics-based detection logic proves vulnerable when confronted with active human modifications, rendering it both unreliable and impractical to rely on these tools for upholding academic integrity at the technical level.

5. Discussion

5.1. Summary of key findings

This study conducts a large-scale empirical evaluation of the effectiveness, fairness, and robustness of 13 mainstream AIGC detection tools in real-world educational settings. Experimental results based on three datasets (StuTask, StuThesis, and DataCode) indicate that while some tools demonstrate a certain level of identification capability in specific genres (e.g., long academic papers), overall, the existing detection ecosystem faces failure when confronted with complex, real student assignments—especially engineering code and polished texts. This indicates the necessity of adopting differentiated assessment strategies tailored to various types of assignment content.

Regarding RQ1, the primary finding of this study reveals significant variations in detector performance across different evaluation scenarios. While commercial tools such as Scribbr.com and GPTZero.me achieved high accuracy in long academic papers (StuThesis), their performance in engineering code and short assignments was disappointing—with some even falling below random guessing. This result strongly contradicts the marketing narratives of commercial companies claiming “all-purpose detection capabilities” and aligns with the conclusions of a multi-center study by [Weber-Wulff et al. \(2023\)](#), which states that “no universal detection solution adaptable to all academic genres currently exists”.

The low detection rate observed in the DataCode dataset of this study validates the assertion by [Pan et al. \(2024\)](#) that existing natural language processing (NLP)-based detectors struggle to transfer to the programming language domain. There exist inherent differences between code and natural language. The detection of natural language texts primarily relies on statistical features such as “Perplexity” and “Burstiness”, assuming that AI-generated texts exhibit lower perplexity and smoother vocabulary distribution. However, programming languages are inherently high-structured and low-entropy. In engineering education, students are trained to write “efficient, concise, and standardized” code, resulting in a high degree of overlap in statistical features between high-quality human-written code and AI-generated “optimal solutions”. As noted by [JianWang et al. \(2024\)](#), the syntactic rigidity of code eliminates stylistic differences between humans and machines, leading statistical regularity-based detectors to fall into “blindness”. This finding carries significant warning implications: in computer science and engineering disciplines, relying on existing tools to determine assignment plagiarism is highly likely to result in large-scale false negatives, thereby condoning academic misconduct.

Furthermore, the performance gap between commercial detectors (e.g., Originality.ai) and open-source models (e.g., Fast-DetectGPT) has sparked a crisis of trust in “black-box algorithms”. While commercial tools perform better on long texts, their excessively high false positive rates (FPR > 40% in StuTask) align with [Yeo \(2023\)](#) warning: due to the lack of transparent confidence thresholds and training data disclosure, educators cannot assess the reliability of detection reports. In contrast, Fast-DetectGPT has

demonstrated superior cross-task generalization capabilities due to its zero-shot mechanism. This indicates that in the absence of domain-specific training data, leveraging the inherent probability curvature of LLMs for detection may represent a more sustainable technical route—rather than relying on opaque commercial APIs.

Regarding the results of RQ2, this study reveals that detectors appear more adept at identifying simple, low-cognitive-load tasks but often fail when confronted with outputs of higher-order thinking. Our finding in StuTask—that the more difficult the task, the harder the detection—can be explained from the perspective of Cognitive Load Theory. Simple tasks typically involve factual recall or basic concept explanation, and LLMs tend to generate textbook-style, highest-probability responses when addressing such tasks; these smooth statistical features are highly distinguishable. However, difficult tasks require complex logical reasoning, multi-step deduction, and in-depth integration. To tackle these challenges, texts generated by LLMs (or high-quality student responses) often incorporate more complex syntactic structures and specialized terminology, where increased linguistic complexity blurs the human-machine boundary. Bearman et al. (2024) note that “AI performance in complex cognitive tasks is increasingly approaching that of human experts, rendering detection based on superficial linguistic features progressively ineffective”. This aligns with the conclusions of our study.

Regarding engineering code, detection results on the DataCode dataset reveal a cliff-like decline in performance as code length increases. Traditional complexity metrics are unable to effectively distinguish machine-generated code from human-written programs (Campbell, 2016). As program length grows, large language models increasingly exhibit strong global structural organization capabilities, generating complete software artifacts with modular decomposition, nested function calls, and explicit exception handling. This form of “structural camouflage” undermines detectors that rely primarily on local token-level distributions, rendering them progressively ineffective for long and complex code.

As a consequence, an unintended asymmetry emerges in academic integrity enforcement. Students face the lowest detection risk when using AI assistance for the most complex and high-stakes programming assignments—typically final projects involving substantial codebases—whereas the use of AI for simpler, low-level exercises or gap-filling tasks is more likely to be detected. This phenomenon constitutes a clear “reverse selection” mechanism that runs counter to the original intent of educational assessment. It systematically shifts detection risk away from high-level integrative problem-solving toward formative and foundational tasks, thereby distorting students’ learning incentives in unintended ways. More fundamentally, this pattern exposes a core limitation of output-based code detection approaches. Effective evaluation in programming education cannot rely solely on surface-level statistical signals in generated code; instead, it necessitates a transition toward process-oriented assessment strategies that foreground students’ reasoning processes, design rationales, and development trajectories.

Regarding RQ3, the findings indicate significant disparities in detector performance across disciplines: STEM texts are high-risk areas for both false positives and false negatives, while humanities and social sciences texts are relatively low-risk. In their landmark study, Liang et al. (2023) note that detectors exhibit severe bias against non-native English speakers (NNES), as their writing tends to have lower vocabulary richness and inherently lower perplexity. This study extends this conclusion to the level of “disciplinary language”. Academic writing in engineering and natural sciences emphasizes objectivity, precision, and passive voice, and this highly formulaic writing style is statistically indistinguishable from AI-generated text. For instance, when describing experimental procedures or mathematical derivations, human students have little choice but to use fixed sentence structures (e.g., “The solution was heated to...”). For commercial detectors trained on general-purpose corpora, such texts lacking stylistic variation are classified as “machine-generated”. This is a finding with significant ethical implications: top science and engineering students who adhere to disciplinary norms and strive for precision face a much higher risk of misclassification than their humanities counterparts. This is not only unfair but also a potential penalty for upholding academic standards.

The robustness test results of RQ4—particularly the AI evasion rate as high as 88% under combined strategies—reveal that there are currently almost no text detectors capable of withstanding intentional human modifications. This finding is highly consistent with Sadasivan et al. (2023) research on “adversarial rewriting”. The core logic of detectors lies in identifying the specific lexical selection probability chains remaining in AI-generated texts. However, this study confirms that simple synonym replacement or sentence restructuring is sufficient to disrupt these fingerprints. Sadasivan et al. refer to this as the “Laundering Effect”: by introducing minor perturbations, students effectively “pull back” the probability distribution of AI-generated text into the distribution range of human writing. While institutions like Turnitin claim to detect “AI-rewritten content”, their actual performance falls short. When students use tools or manually polish texts, they are essentially using human linguistic habits to break machine-generated patterns. Combined modification strategies erase machine traces across multiple dimensions, leading to a decline in detection rates. As Krishna et al. (2023) notes, as long as the distribution of generative models can be approximated to human distribution through simple post-processing, statistically based passive detection is theoretically unfeasible. This indicates that relying solely on technical means to uphold academic integrity is a destined-to-fail “cat-and-mouse game”.

5.2. Implications

Given that this study reveals the instability of detection tools in STEM disciplines and their extreme vulnerability to adversarial attacks, the higher education sector must recognize that relying solely on “technical containment” not only fails to curb AI cheating but also creates new forms of academic injustice. Under these limitations, the role of detectors should increasingly shift toward serving as “supplementary indicators for assessment”. Against the backdrop of higher education institutions increasingly viewing Generative Artificial Intelligence (GenAI) as an integral component of content development and assessment design, the empirical results of this study offer multi-dimensional insights for the transformation of educational assessment systems.

At the level of policy formulation and assessment governance, universities should move beyond a narrow paradigm of *outcome monitoring* toward a model of *educational diagnosis*. The high evasion rates identified in this study, together with the pronounced algorithmic blind spots of existing detectors in STEM disciplines, demonstrate that treating detection reports as the sole “smoking gun” for academic misconduct is both technically unreliable and educationally hazardous. Accordingly, the function of AIGC detectors should be reconceptualized—not as compliance inspectors, but as one diagnostic signal among multiple complementary indicators within an interdisciplinary evaluation framework.

This shift is particularly critical in STEM and programming contexts. In these domains, high-quality human-written code and AI-generated optimal solutions often converge in logical structure, modular organization, and syntactic regularity. As a result, detectors that rely primarily on statistical surface features are inherently limited in their ability to distinguish between rigorous human reasoning and advanced AI assistance. Rather than serving as binary arbiters of misconduct, detectors in STEM settings should be repositioned to estimate the extent of AI involvement in standardized components and logical synthesis. This reframing enables assessment practices to redirect attention toward more pedagogically meaningful evaluations, such as oral defenses of core algorithmic principles, step-by-step justification of design choices, and in-depth verification of students’ problem-solving trajectories.

As emphasized by White (2023), the imposition of penalties without interpretability and due process violates fundamental principles of educational ethics. In line with this view, institutional policies should explicitly prohibit penalizing students solely on the basis of automated AI detection outcomes in the absence of corroborating evidence. To mitigate the risks posed by algorithmic bias—particularly for STEM students and non-native English speakers—universities should establish hybrid review mechanisms that triangulate detection results with version histories, process artifacts (e.g., drafts or mind maps), and oral examinations.

In humanities and social science writing, detectors should serve as indicators of “narrative patterns” to identify formulaic outputs lacking unique perspectives, guiding the evaluative focus back to students’ critical argumentation in complex contexts and the integration of personal embodied experiences. In arts and creative disciplines, their relevance lies in capturing the degree to which a work breaks through standard academic paradigms, focusing on the dynamic interplay between AI style simulation and human mastery of aesthetic innovation and deep intent. Through these discipline-specific indicator settings, educators can more precisely map the boundaries of human–AI symbiosis within different knowledge domains, transforming technological feedback into a scientific basis for optimizing pedagogical design and enhancing students’ AI literacy.

At the level of instructional tasks and assessment design, educators need to shift from “preventing AI” to “integrating AI”. This study found that as the cognitive complexity of a task increases, the detector’s ability to identify AI weakens—as evidenced by the significant decline in AUC for “Hard” tasks—suggesting that traditional assignments based on factual description or standardized application are no longer viable. Consistent with the “embrace rather than avoid” philosophy advocated by Novianity et al. (2024), teachers should proactively integrate AI into cognitive scaffolding. The focus of assessment design should shift toward the “evaluation” and “creation” dimensions at the top of Bloom’s Taxonomy (Lodge et al., 2023). By designing strongly situational and embodied tasks integrated with real-time instructional interaction—such as in-class debates or local community surveys—educators can enhance the “anti-forgery” capacity of tasks. This approach not only circumvents the detection dilemma but also effectively cultivates students’ “AI literacy”.

Finally, the assessment system must complete a radical transition from an “outcome-oriented” to a “process-oriented” approach. The quantitative analysis of adversarial editing behaviors in this study proves that static detection targeting final texts is theoretically an unsustainable “cat-and-mouse game”. Lodge et al. (2023) note that this continuous tracking of thinking trajectories is the most powerful evidence to prove knowledge internalization and a deep cognitive process that AI struggles to simulate. Therefore, the ultimate line of defense for safeguarding academic fairness lies in making the learning process “visible”. Higher education institutions should promote longitudinal assessment systems, requiring students to submit an “evidence chain” consisting of topic selection proposals, records of multiple draft iterations, and critical modification journals regarding AI-generated content. For high-risk domains such as code generation, synchronized interpersonal interaction assessments, such as interactive oral examinations, should be introduced to verify students’ actual mastery of logic through real-time questioning.

Regarding AIGC detectors themselves, based on the vulnerability of existing tools in addressing adversarial modifications as revealed by this study, future AIGC detectors must undergo an architectural optimization shifting from “sentential features” to “logical innovation”. Current detection logic relies excessively on superficial statistical features such as perplexity, which often penalizes high-quality human work characterized by linguistic rigor and clear logic. Future optimization should be dedicated to identifying “cognitive increments” within a text—specifically, whether the work demonstrates a unique stance, complex interdisciplinary associations, or logical innovations that transcend the standardized training distributions of models. Optimized tools should assist educators in identifying how students have enhanced the depth of their thinking through AI assistance, rather than obsessing over the origins of sentential expressions. Such a transformation will align detection technology more closely with an “AI-embracing” academic environment, evolving it from a defensive containment tool into an educational support system that fosters the development of AI literacy and evaluative judgment. This will further facilitate the advancement of higher education assessment reform and establish a line of defense for academic integrity that is truly adapted to the future educational ecology.

5.3. Limitations and suggestions

Despite the scale and scope of this study, several limitations should be acknowledged. Firstly, although the data span multiple academic years, they are primarily drawn from a single university, which may introduce institutional and geographical bias. Secondly, the AI-generated content evaluated in this study is mainly produced by GPT-3.5. With the rapid adoption of more advanced generative models, such as GPT-4 and Claude 3, the difficulty of reliable detection may further increase, potentially

altering the relative performance of existing tools. Thirdly, a notable limitation is that all experimental datasets—StuTask, StuThesis, and DataCode—are grounded exclusively in English academic content. While English currently dominates both AIGC development and higher education research, different languages exhibit substantial variations in linguistic structure, rhetorical conventions, and the distribution of model training data. As a result, the statistical regularities leveraged by current detection algorithms in English contexts (e.g., perplexity and related metrics) may face new challenges in terms of effectiveness and robustness when generalized to other languages, such as Chinese or Spanish.

Future research should therefore extend this work in several directions: (1) exploring multimodal detection approaches that integrate textual features with writing-process data; (2) conducting longitudinal studies to examine the long-term co-evolution of student writing behaviors and AI usage; and (3) systematically evaluating the cross-linguistic generalizability of AIGC detectors across diverse language settings.

6. Conclusion

This study demonstrates that current AIGC detection tools are not yet sufficiently robust or reliable for high-stakes academic decision-making. The tide of generative AI is irreversible—it is both a disruptor and an innovator. Through conclusive empirical data, this study demonstrates that current AIGC detection tools, while achieving a certain degree of accuracy in judgments, fail to provide substantive guarantees and may even introduce new injustices due to misclassifications. The future of higher education lies not in building higher technical walls, but in redefining the essence of education. We need to construct a new academic contract: one in which integrity stems not from fear of being caught, but from respect for the process of knowledge creation; and assessment is not a mechanical judgment of outcomes, but an affirmation of the unique value of human thinking. In this new era of human–AI symbiosis, only by returning to the true essence of education—nurturing critical thinking, creativity, and ethical judgment—can we establish the ultimate line of defense for upholding academic integrity.

CRedit authorship contribution statement

Yicheng Sun: Writing – original draft, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yihan Liao:** Visualization, Validation, Methodology, Investigation. **Xiaoxue Ma:** Writing – review & editing, Supervision, Project administration, Investigation, Data curation.

Funding

This research is funded by the Hong Kong Metropolitan University R&D Fund (Grant No. RD/2025/1.24). All authors gratefully acknowledge the financial and academic support provided by this institution.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to express their sincere gratitude to the editor and the anonymous reviewers for their time, effort, and valuable feedback on this manuscript. Their constructive comments and insightful suggestions have greatly improved the clarity, rigor, and overall quality of the paper. The authors also wish to thank all participants and contributors who generously devoted their time and effort to this study. Their support and involvement were essential to the completion of this research.

Data availability

Data will be made available on request. Requests for access should be directed to Mr. Yicheng Sun, subject to ethical approval and data protection requirements.

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